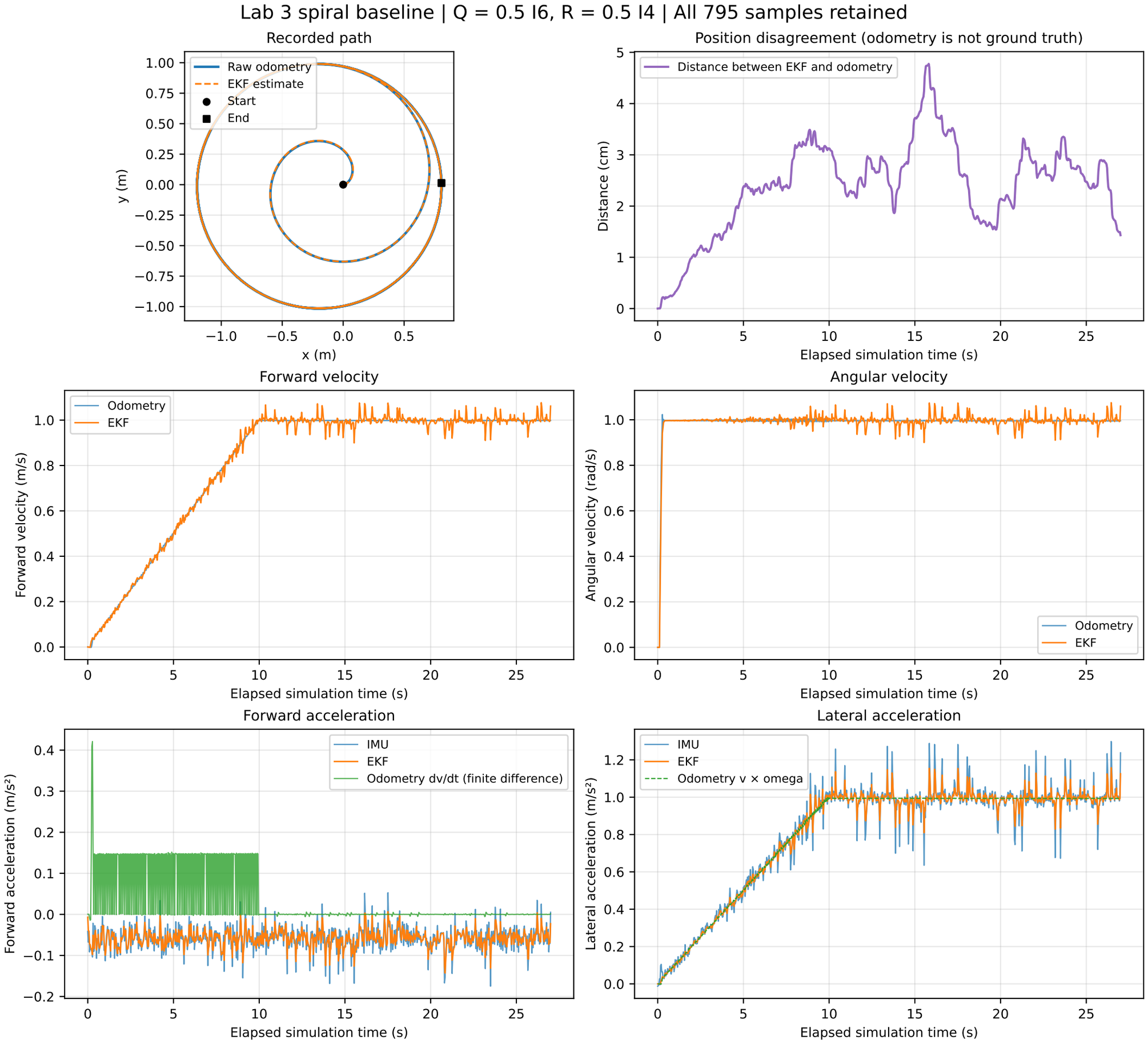

Lab 3 spiral baseline | Q = 0.5 I6, R = 0.5 I4 | All 795 samples retained
Recorded path
Raw odometry
EKF estimate
Start
End
x (m)
y (m)
Position disagreement (odometry is not ground truth)
Distance between EKF and odometry
Distance (cm)
Elapsed simulation time (s)
Forward velocity
Odometry
EKF
Forward velocity (m/s)
Elapsed simulation time (s)
Angular velocity
Odometry
EKF
Angular velocity (rad/s)
Elapsed simulation time (s)
Forward acceleration
IMU
EKF
Odometry dv/dt (finite difference)
Forward acceleration (m/s²)
Elapsed simulation time (s)
Lateral acceleration
IMU
EKF
Odometry v × omega
Lateral acceleration (m/s²)
Elapsed simulation time (s)